

Communication Modules for IoT

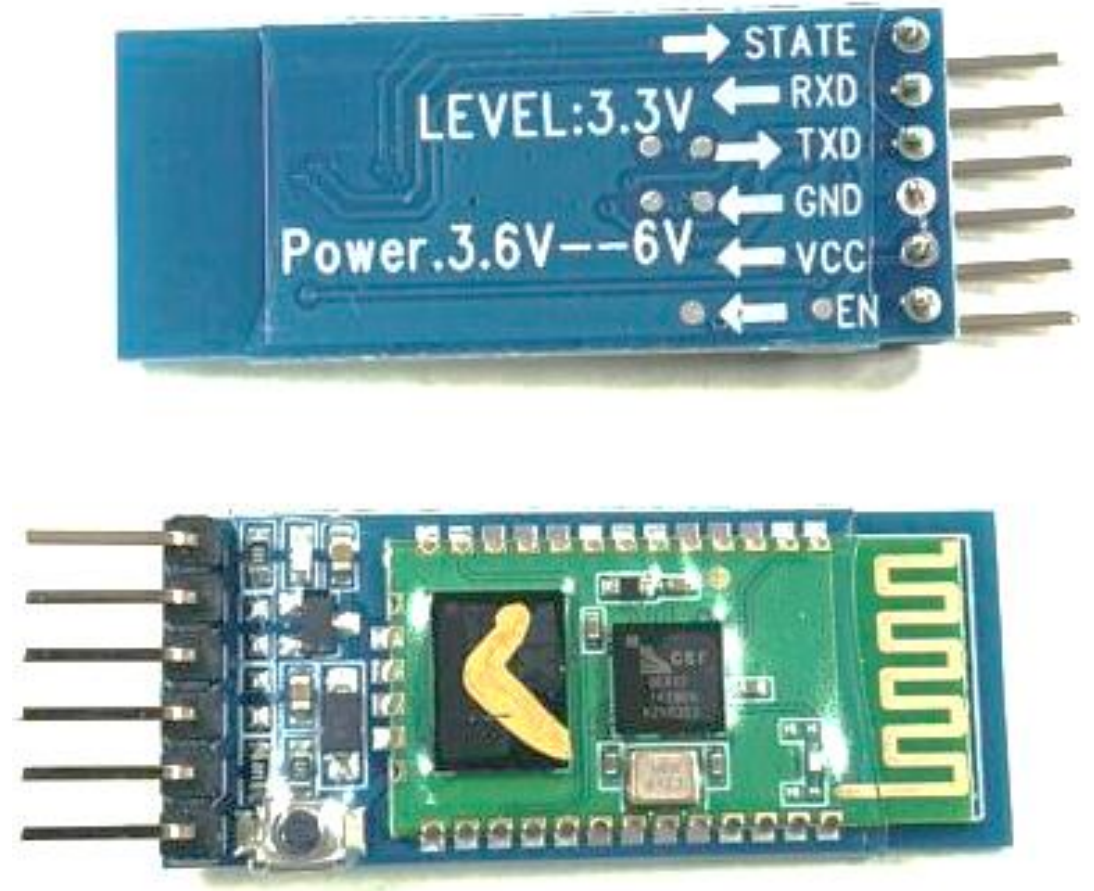
Wi-Fi

- A Wi-Fi module is a small, embedded electronic component that enables devices—such as IoT sensors, smart home appliances, and microcontrollers—to connect wirelessly to the internet or other network devices using 802.11 standards.
- Wi-Fi module is a hardware device that integrates Wi-Fi communication functions, usually including Wi-Fi chips, antennas, connection and control circuits.
- WiFi (wireless LAN) networks use radio waves to communicate between devices.



Bluetooth

- Bluetooth is a short-range wireless technology standard that is used for exchanging data between fixed and mobile devices over short distances and building personal area networks (PANs).
- In the most widely used mode, transmission power is limited to 2.5 milliwatts, giving it a very short range of up to 10 metres (33 ft). It employs UHF radio waves in the ISM bands, from 2.402 GHz to 2.48 GHz.
- The IEEE standardized Bluetooth as IEEE 802.15.1



LoRa

- LoRa (short for “Long Range”) is a radio modulation technology for wireless LAN networks in non-cellular Low Power Wide Area (LPWA) wireless communication network technologies.
- The physical (PHY) silicon layer, or wireless spread spectrum modulation technique, creates a non-cellular, long-range communication link.
- These modules support global ISM bands from 868 MHz to 916 MHz and offers transmission distance of 2-5km in towns and 10-15km in suburbs or rural areas.



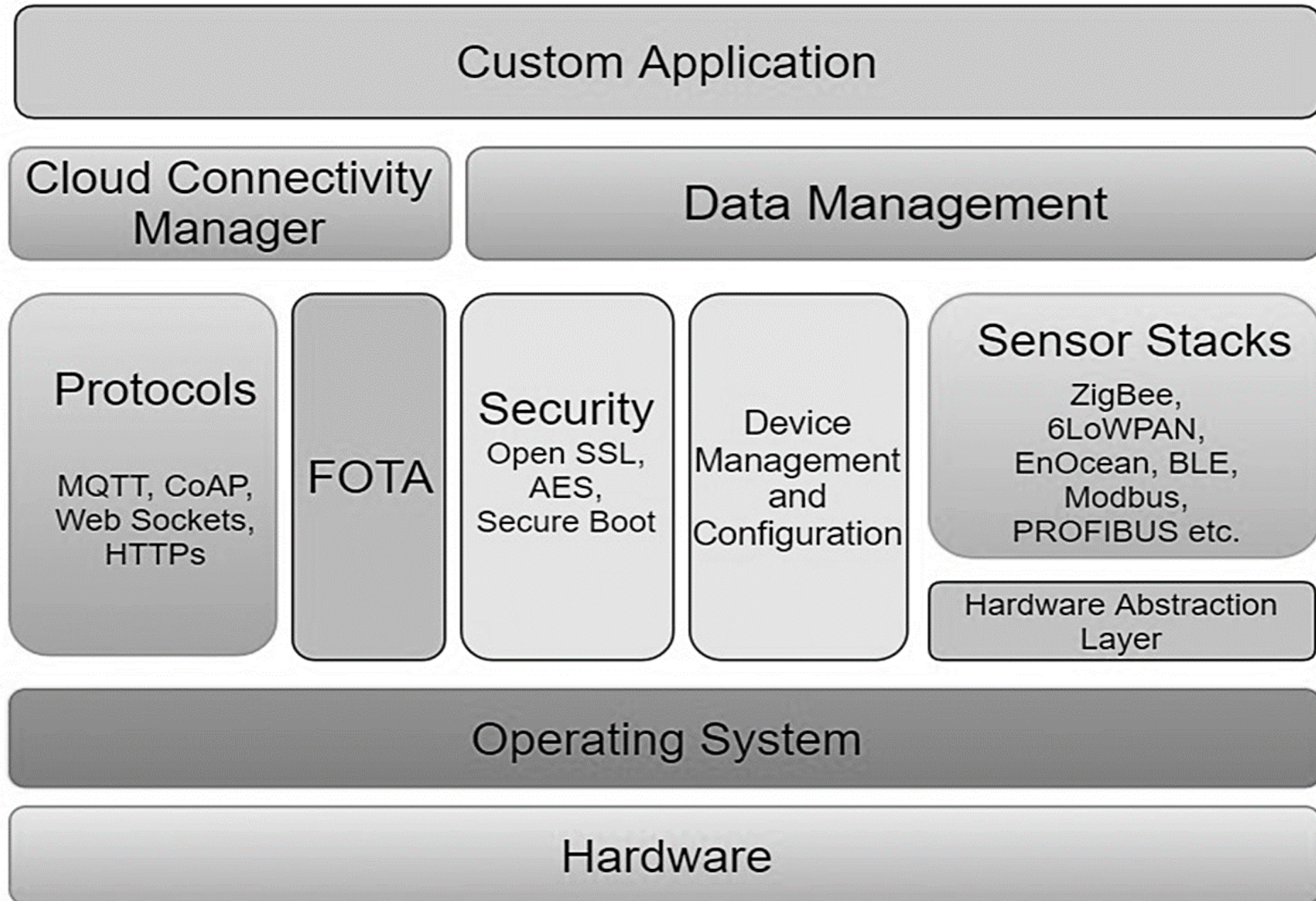
ZigBee

- **Zigbee** is an IEEE 802.15.4-based specification for a suite of high-level communication protocols used to create personal area networks with small, low-power digital radios.
- Its low power consumption limits transmission distances to 10–100 meters (33–328 ft).
- Zigbee devices can transmit data over long distances by passing data through a mesh network of intermediate devices to reach more distant ones.
- Zigbee has a defined rate of up to 250 kbit/s for the 2.4 GHz band



IoT Gateway

- The gateway converts information, data or other communications from one protocol or format to another.
- An Internet gateway can transfer communications between an enterprise network and the Internet.



Device Hardware - IoT Gateway Hardware comprises of processor/microcontroller, IoT sensors, protection circuitry, connectivity modules (e.g Zigbee, Bluetooth, WiFi and more).

HAL (Hardware Abstraction Layer) Hardware Abstraction Layer supports reusability and portability of the IoT software. This layer makes the software design independent of the underlying hardware platform.

Operating System - If the gateway is to be designed for a simple to medium scale application then a RTOS is used; however if the gateway has to perform considerably complex operations then Linux is preferred.



IoT Sensors Stack - This layer basically consists of software stacks that serve as interfaces with IoT sensors modules. Specific stack(s) is/are integrated depending on the sensor interface the IoT Gateway has to support.

Device Management and Configuration - An IoT gateway needs to interface with different types of Sensor devices. The configuration and settings of all the IoT Sensor Devices is stored in the gateway device memory. This ensures that the last saved settings are available after every re-boot.

Security - Secure boot is implemented to ensure that the gateway doesn't boot from an unauthorized firmware. All messages between gateway and cloud, and messages between Gateway sensor node is encrypted to ensure integrity, and confidentiality of sensor nodes.

